Predictive maintenance on EDFA & SFP telemetry

AI/ML Engineer II take-home · Commtel Networks

Runs end-to-end from the raw files in 93 s on a 4-core laptop CPU

The short version

Most of what looks like a fault in this data is the monitoring system, not the equipment

22/88

days with EDFA optics

Collector gaps of 23 and 21 days hit every asset at once

0

device faults confirmed

Every alert traces to a shift with a monitoring signature

24/24

injected ageing ramps caught

0 / 24 alerts on explained hot afternoons

16 wk

SFP lasers to 70 °C

At the Jan–Mar warming rate (70 °C limit assumed)

Approach: context-aware physics residuals (cooler | case temperature, pump current | input, pump power per mA, pump temperature | set-point), robust z-scores with physical floors, 15-minute persistence. Validated with injected faults, an Isolation Forest cross-check and an independent data-integrity rule.

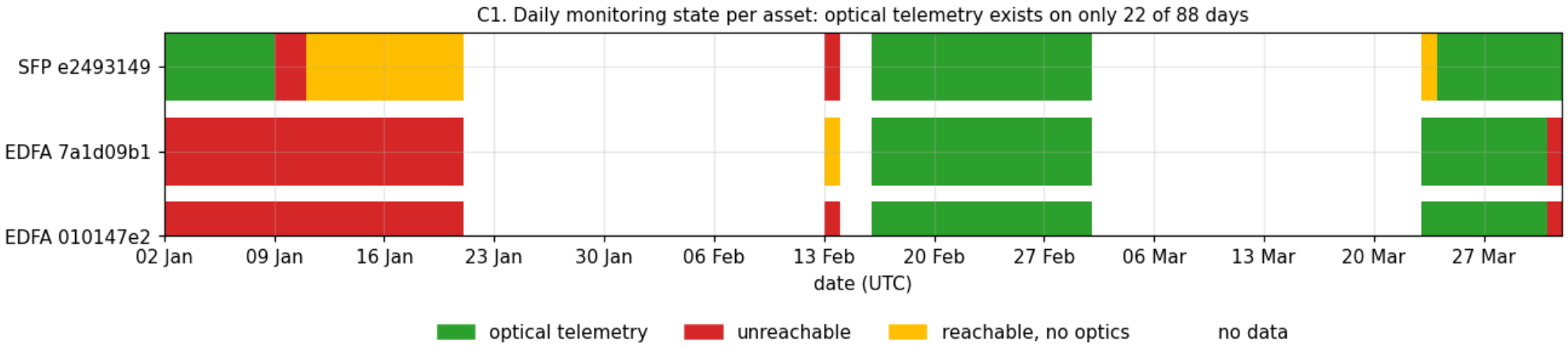
What arrived, and how it was shaped

5.58 M long-format readings → one row per asset per 1-minute polling slot, every sensor classified

1 **Kept the 41% nobody mapped**
60 SFP ids, 1.78 M rows: kept as UNMAPPED_<id>, role inferred from values (−48 V feeds, chassis temperature)

2 **Aligned on 1-minute slots**
Sensor groups poll at :13, :21–33, :51, :53. Clustering by gap would chain them, so floor to the minute and count collisions

3 **Classified from data, not names**
Pump2 = junk (single-pump units), Uptime (hardware) = 0, Current: Timestamp = 0 sentinel; 130 k duplicate rows removed



Preprocessing decisions

Every rule is logged with what it touched; nothing is dropped silently

1. **Outages masked, never filled**
   ICMP loss 100% (+5 min guard): 51 k EDFA asset-minutes

2. **Impossible values → point anomalies**
   4 EDFA sentinels (1.0 mA, 1.0 mW); 30,859 SFP incl. 448 zero voltages

3. **SFP window = Current**
   Only window with no impossible values that is aligned to poll time

4. **5 min EDFA, 15 min SFP medians**
   Per-asset scaling: the two EDFAs run at 15 and 7 dBm

Share of SFP readings outside the physical envelope

Metric	Current	Prev 15min	Current Day	Prev Day
Laser Voltage	0.0%	0.0%	**23.8%**	0.0%
Laser Bias	0.0%	0.0%	**1.8%**	0.0%
Transmit Power	0.0%	0.0%	**1.0%**	0.0%
Laser Temperature	0.04%	0.03%	**0.5%**	0.0%
Received Power	0.0%	0.0%	**0.3%**	0.0%

Previous 15min also runs 49–94 min ahead of the poll clock, so Current is the only usable 15-min view; the day windows stay as a cross-check.

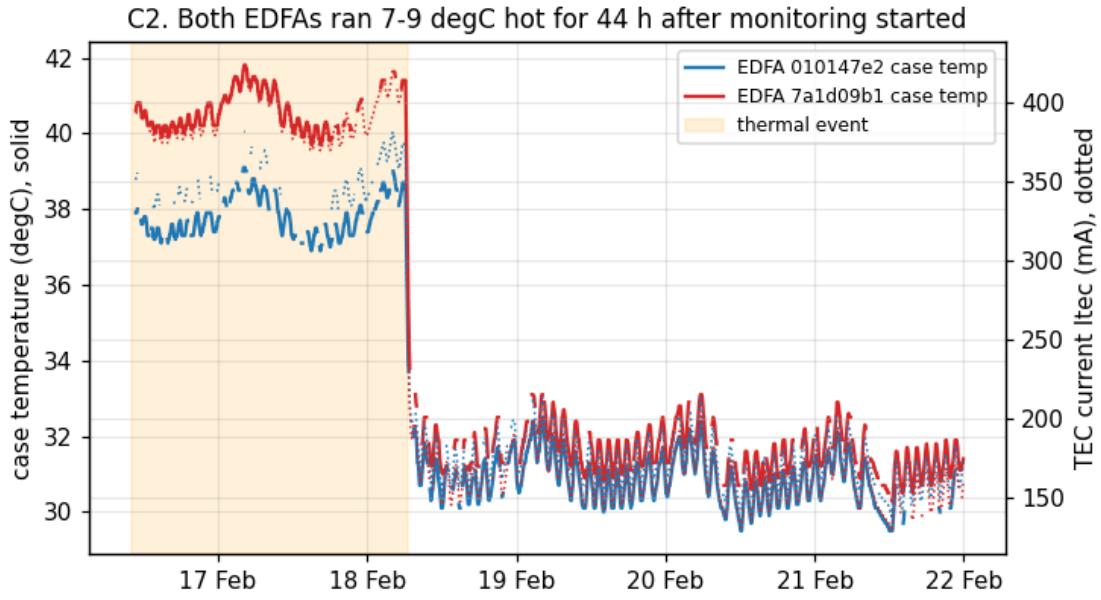
A heat event that was the site, not the EDFAs

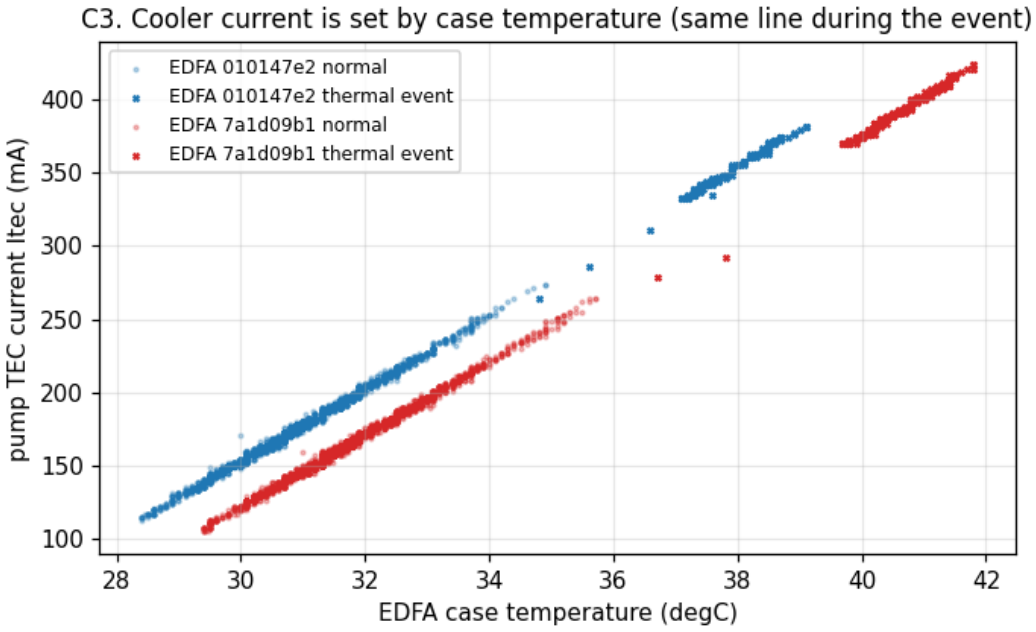
16 Feb 10:41 – 18 Feb 06:30 UTC: both units hot together, coolers handled it

1 **+7–9 °C on both units at once**
Case 37.7 / 40.4 °C vs 30.8 / 31.6 °C after; the two units' temperatures correlate at r = 0.995

2 **Cooler current ×2, pump held 25 °C**
Itec 349 / 386 mA vs 173 / 161 mA; pump chip never left its set-point

3 **Itec = 25 mA per °C of case**
r = 0.997 on both units; the room is warmest 02:00–05:00 IST (scheduled HVAC?)





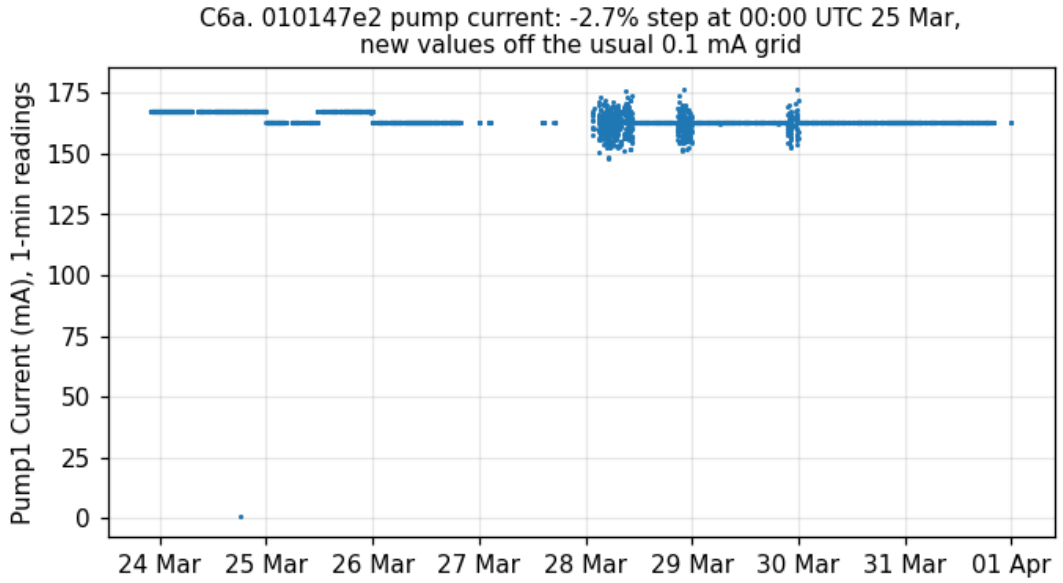
Pump faults the physics does not support

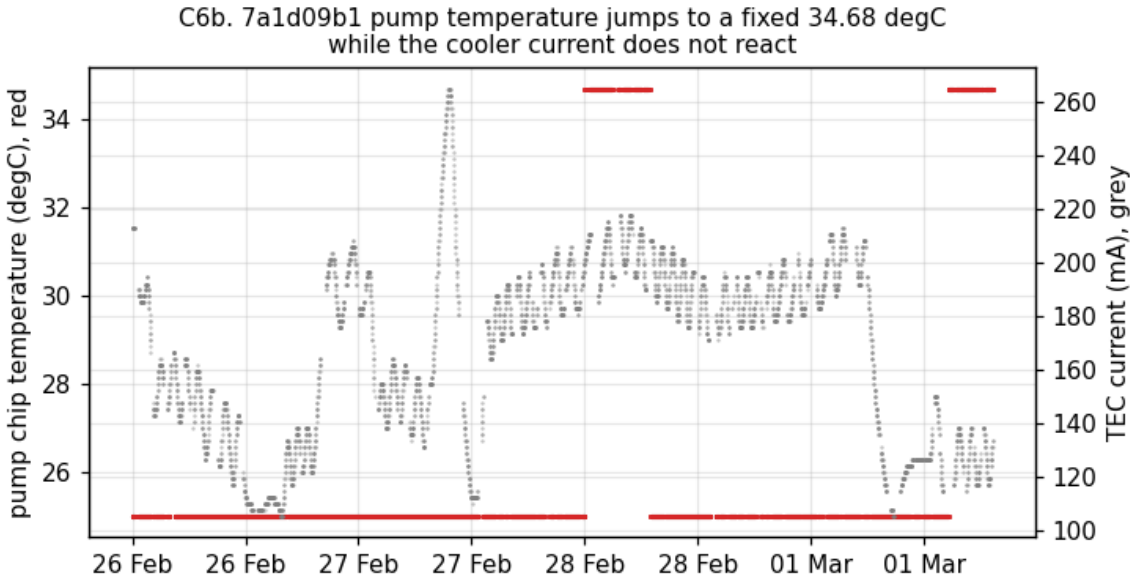
Three EDFA shifts start at exactly 00:00 UTC with values the sensor never produced before

1. **7a1d09b1 pump power −28%**
   26.0 → 18.6 mW on 22/24 Feb; current and POUT flat. A real pump loss would sag POUT

2. **010147e2 pump current −2.7%**
   167.2 → 162.67 mA from 25 Mar, off its 0.1 mA grid. Ageing pumps need more current, not less

3. **7a1d09b1 pump temp stuck**
   Fixed 34.68 °C (+9.7 °C) while the cooler current does not react at all





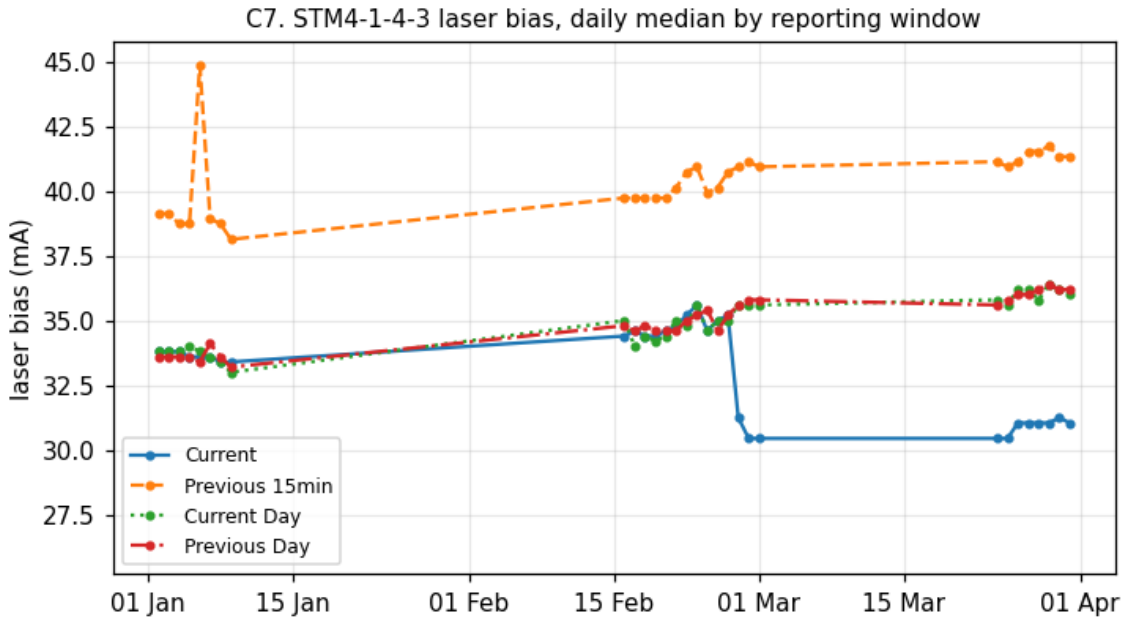
SFP reporting windows disagree

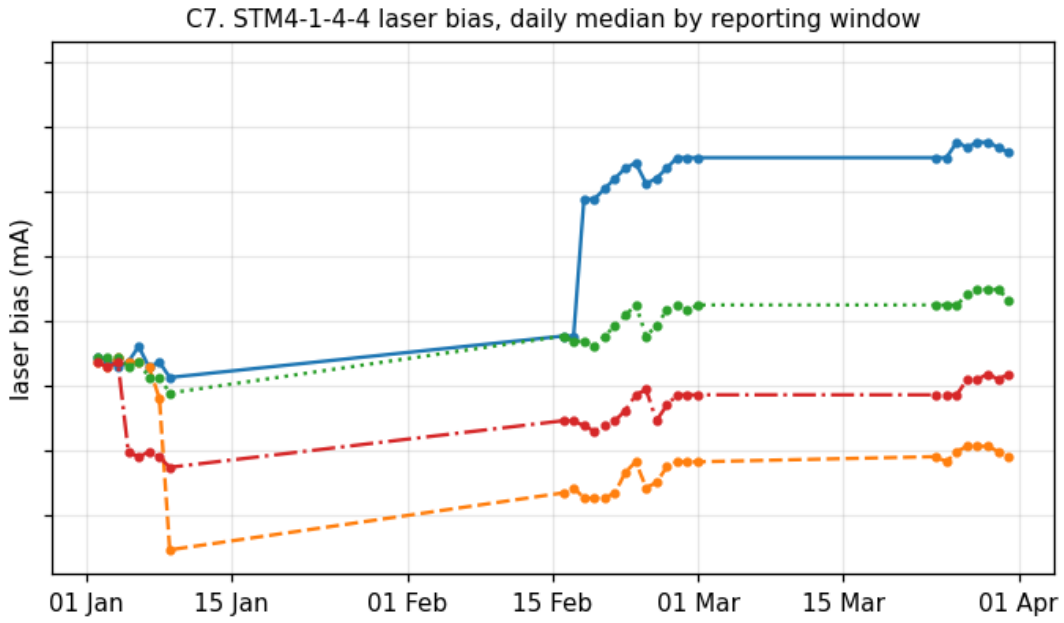
Every SFP step change lives in a single window, so it is the reporting path, not the laser

1 **Current steps alone**
1-4-3 Tx −12% (6 Jan) · 1-4-4 bias +16% (18 Feb) · 1-4-3 bias −4.4 mA (27 Feb); day windows flat

2 **Current Day is corrupted**
23.8% impossible voltages, negative bias and Tx; 1-4-3 Rx 2–6 dB off

3 **Previous 15min clock is fast**
Window stamps 49–94 min ahead of the poll; bias offset +5.3 mA on 1-4-3. No port flaps in 88 days





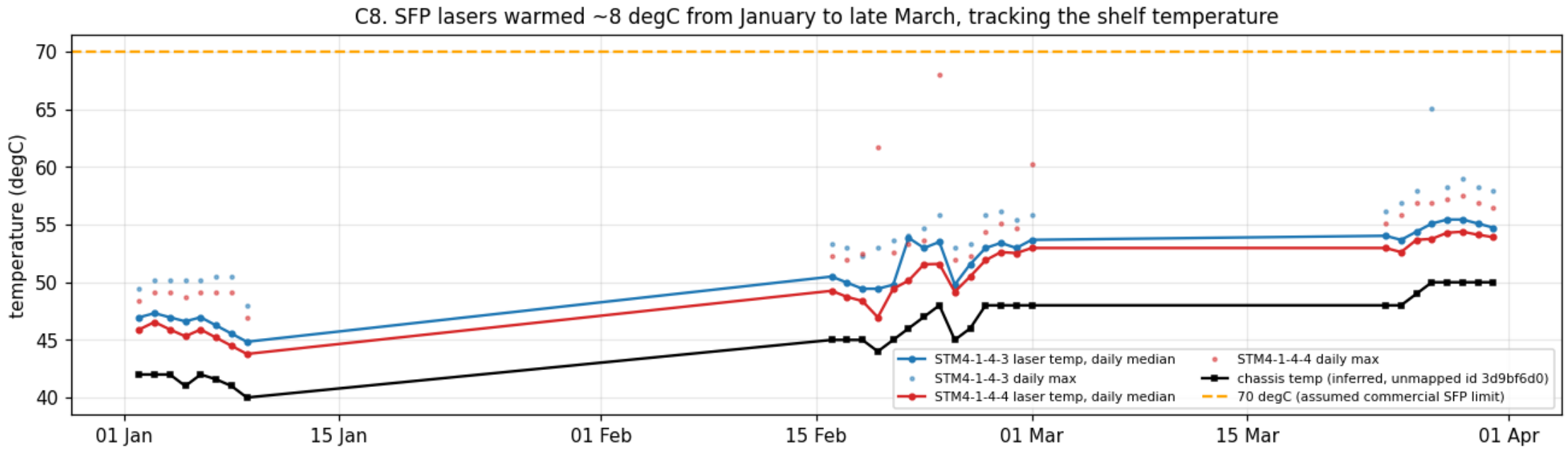
The near-term SFP risk is heat, not ageing

Both lasers warmed ~8 °C from January to late March, tracking the shelf

1 **0.73 °C per week**
46.6 → 55.0 °C and 45.6 → 53.9 °C; late-March daily peaks 58.1 / 56.9 °C

2 **~16–18 weeks to 70 °C**
Mid-July at this rate, before the Indian summer adds more (70 °C commercial limit assumed)

3 **Bias rise is physics**
+0.29–0.31 mA per °C fitted; inferred shelf sensor correlates r = 0.81 / 0.92



Detector: four physics questions

Contextual residuals of 1–3-coefficient models; one robust z-score per question

1 Cooler

Itec given case temperature

≈ 25 mA/°C, R^2 0.99

floor 3 mA

2 Pump drive

Current given PIN and temperature

POUT pinned by the loop

floor 0.8 mA

3 Pump efficiency

Pump power per mA

primer's degradation signature

floor 0.01 mW/mA

4 Pump temp

Chip vs its 25 °C set-point

lost thermal regulation

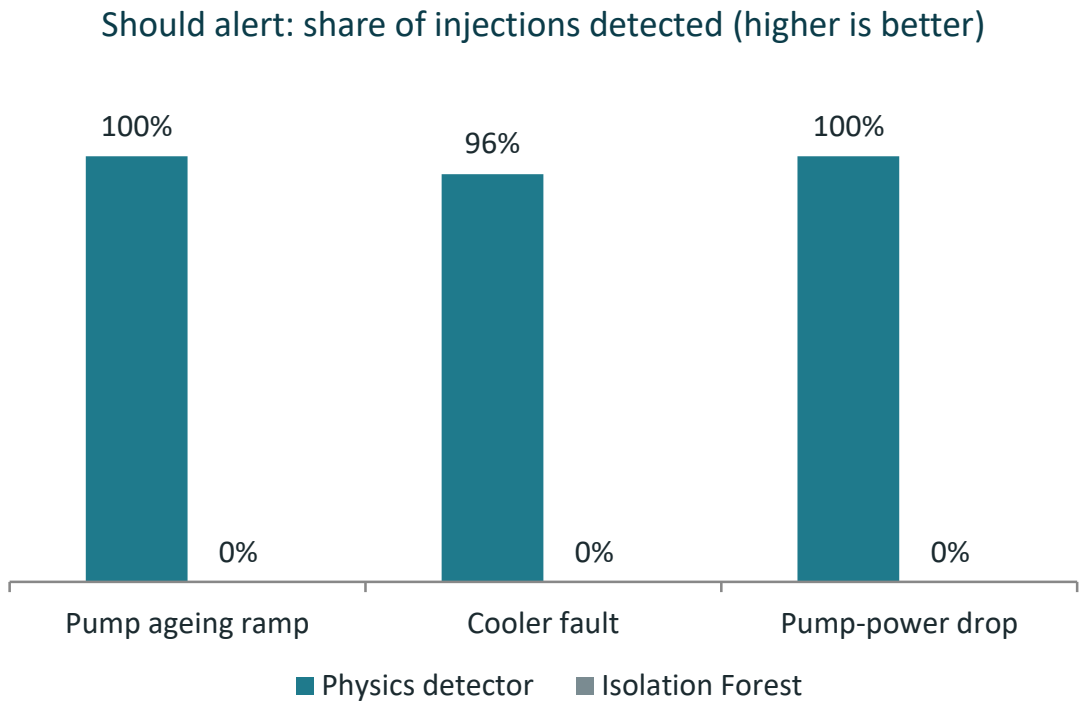
floor 0.25 °C

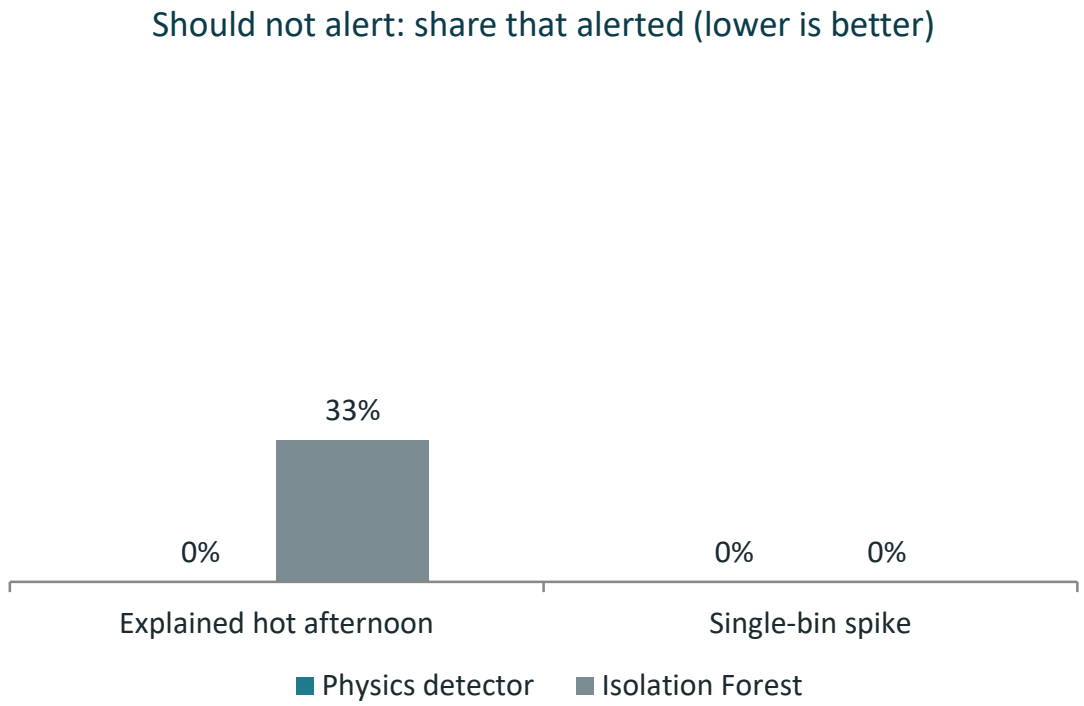
Alert rule |z| > max(4, training 99.9th percentile) for 15 min · incident stays open until clear for 6 h · trained 18–21 Feb: after the heat event, no outages, before the first shift

Rejected: autoencoder / LSTM (no labels, ~22 days per unit, not explainable) · raw thresholds (fire every warm night) · Isolation Forest as the main detector (evaluated on the next slide). Scoring costs a few multiply-adds per bin on any CPU.

Evaluation without labels

12 random placements per fault per unit, frozen models; only new alerts caused by the injection count

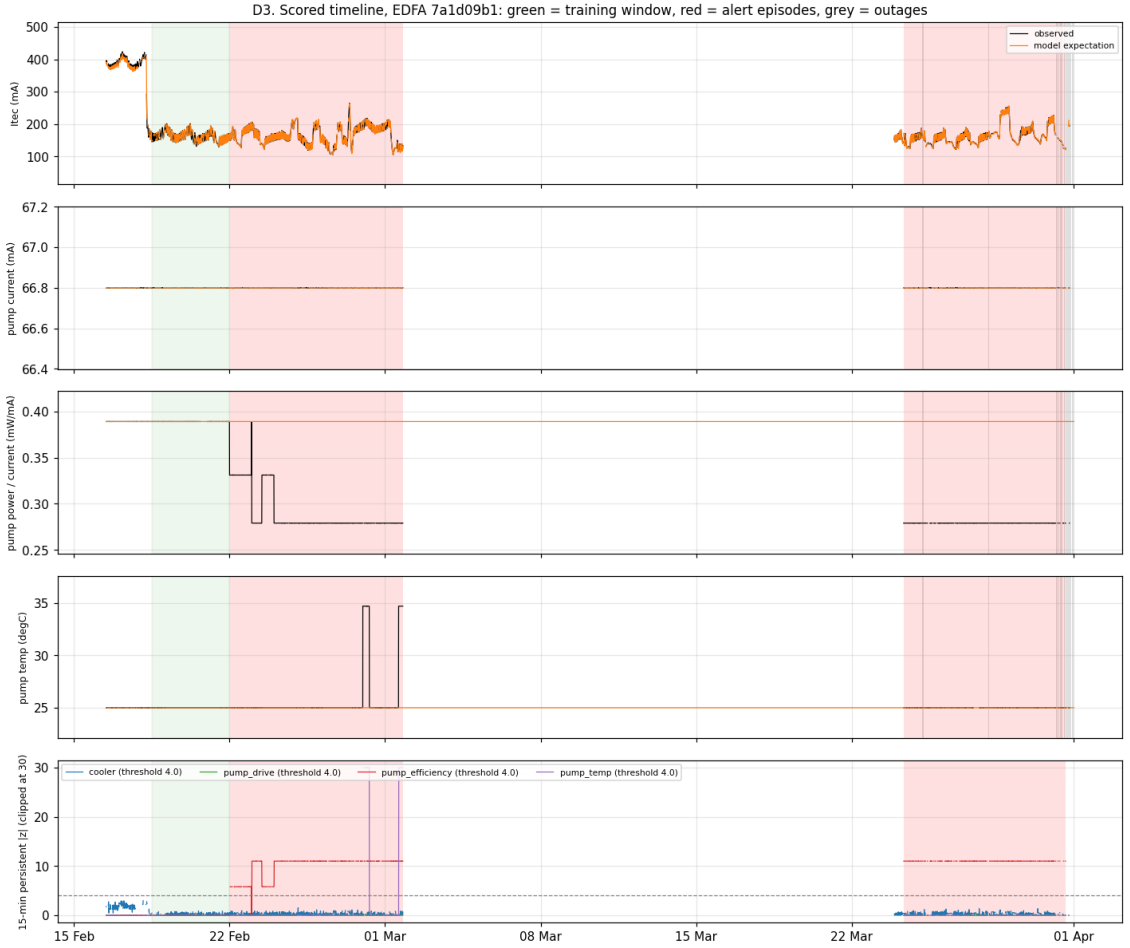




Ageing ramps alert at ~2% drift on 010147e2. The forest cannot split on a signal that was constant in training (7a1d09b1 pump power), and it alerts on variation that context explains. That contrast is why the framing is contextual.

What it flagged on the real data

Zero alerts in training or on the cooler model; three root causes, all judged monitoring-side



Asset · signal	Period (UTC)	Verdict
7a1d09b1 · pump power −28%	22 Feb – 1 Mar, 24–31 Mar	Monitor / reporting path; verify on site
7a1d09b1 · pump temp 34.68 °C	28 Feb, 1 Mar (hours)	Stuck value; cooler unaffected
010147e2 · pump current −2.7%	25–31 Mar, on and off	Monitoring artefact, low severity
010147e2 · pump temp −0.007 °C	from 22 Feb	Correctly ignored (below 0.25 °C floor)
SFP · single-window steps	6 Jan, 18 Feb, 27 Feb	Reporting path; day windows flat

No gradual trend exists to extrapolate: every slope is produced by a single step, so no lead time is claimed.

Deploying it on site

The model is the smallest part of the system

Deployment sketch

1. **Every 5 minutes**
   pandas/numpy job beside the NMS poller; writes incidents tagged device or monitoring
2. **Monthly refit**
   or on hardware, firmware or set-point change (visible in config sensors)
3. **Feedback**
   engineer closes each incident real / explained / false; false windows leave the next refit
4. **Compute**
   1 vCPU, 1 GB RAM per 100 assets; no GPU, no internet

With 10× the data and two more weeks

1. **A year of gap-free data**
   seasonal cooler model, real ageing slopes, honest lead times
2. **Vendor limits**
   pump EOL current, TEC max, SFP part numbers: z-scores become margins
3. **Tickets as weak labels**
   calibrate floors and verdicts
4. **Data-quality monitor**
   window consistency, quantisation breaks, collector gaps, reported separately